

[Quiz—2] Spring 2024: CSE 309: Theory of Automata

— SOLUTION —

Tuesday, February 27, 2024. Total Marks: 10, Duration: 15 minutes

Question 1

[10 points]

Let L be a language over some Σ^* defined as

$$L = \{w : w = w^{\text{rev}}\}$$

where if $w = w_1w_2 \dots w_n$ then $w^{\text{rev}} = w_nw_{n-1} \dots w_1$ (i.e., the reverse).

Show that L is not regular using the *pumping lemma*.

Solution: Proof by contradiction:

Let us assume L is a regular language. Then, by the pumping lemma, there exists a pumping length p such that any string $s \in L$ with $|s| \geq p$ can be split into three parts $s = xyz$ such that the following conditions hold:

1. $|xy| \leq p$,
2. $|y| \geq 1$, and
3. $xy^iz \in L$ for all $i \geq 0$.

Without loss of generality we can assume $\Sigma = \{0, 1\}$. Now we choose a string $s \in L$. Suppose $s = 0^p10^p$. Clearly, $s \in L$ and $|s| \geq p$. We now split s into three pieces as $x = 0^{p-1}$, $y = 0$, and $z = 10^p$. Clearly, $|xy| \leq p$ and $|y| \geq 1$. Now, we consider $s' = xy^2z = 0^{p-1}0010^p = 0^{p+2}10^p$. Since $p+2 \neq p$ and $s' \neq (s')^{\text{rev}}$, $s' \notin L$. This contradicts the pumping lemma. Therefore, L is not regular.